

In vitro evaluation of effectiveness of antifungal agents against Macrophomina

Phaseolina

Krishna Ramakrishnan Acharya, Aswin Reddy Chilakala and Muthappa Senthil-Kumar

*Corresponding author

Muthappa Senthil-Kumar

National Institute of Plant Genome Research, New Delhi, India-110067

Tel- 91-11- 26735229,

Fax- 91-11-26741658

E-mail- skmuthappa@nipgr.ac.in

<http://orcid.org/0000-0003-1502-1659>

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Abstract

Antifungal agents provide both preventive and curative effects by protecting plant surfaces and eliminating existing infections within plant tissues. Evaluating their impact

against phytopathogenic fungi is critical for sustainable agriculture. *Macrophomina phaseolina* is a highly destructive soil-borne pathogen with a broad host range, causing charcoal rot and related diseases that result in severe yield losses in crops such as soybean, sorghum, maize, and legumes. Several methods are employed to assess the in vitro antifungal efficacy of various substances, including broth microdilution, disk diffusion, gradient screening, and agar-based screening. However, these conventional techniques are generally slower, more labour-intensive, and limited by reproducibility issues. The adoption of microplate reader–based antifungal efficacy testing has enhanced efficiency and scalability, enabling high-throughput evaluation of multiple antifungal agents across a range of concentrations while providing a streamlined platform for preliminary screening. This chapter presents a convenient and cost-effective protocol for assessing the activity of antifungal agents against fungi that produce microsclerotia, with emphasis on standardisation and quality control measures. The antifungal efficacy testing and cell death assay are employed to determine inhibitory concentrations via spectrophotometry and to quantify hyphal death through fluorescence microscopy, respectively. In addition, detailed procedures for inoculum preparation, antifungal agent dilution, incubation conditions, and statistical analysis are described.

Keywords: Phytopathogenic fungi, Antifungal efficacy testing, micro-broth dilution, spectrophotometry, microsclerotia, hyphal death, fluorescence microscopy.

1. Introduction

1.1 *Characteristics of *Macrophomina phaseolina**

M. phaseolina is a necrotrophic fungus that causes stem and root rot, charcoal rot, and seedling blight, particularly under heat and drought stress. It propagates through dark-coloured, branching hyphae and produces microsclerotia during infection [1]. Being highly polyphagous, it infects cereals, legumes, oilseeds, vegetables, fruits, tubers, and weeds across diverse habitats worldwide [2]. Consequently, developing high levels of resistance against *M. phaseolina* remains challenging. The available biological sources of AFAs (antifungal agents) include proteins, secondary metabolites, and chemical compounds derived from plants and microorganisms [1].

1.2 *Emerging Antifungal Efficacy Testing Methods*

The initial evaluation constitutes the screening phase preceding the scalable implementation of the identified candidate AFA. Appropriate assays enable the comparison of dosage requirements and efficacy of the investigated AFAs in terms of growth inhibition. Standard guidelines for antifungal efficacy testing are provided by the European Committee on Antimicrobial Susceptibility Testing and the Clinical and Laboratory Standards Institute [3, 4]. Conventional approaches, such as broth dilution, broth microdilution, disk diffusion, and agar-based screening, are available [5]. However, these methods lack standardisation and are prone to difficulties in visual readout, long turnaround times, and limited applicability across different AFAs and fungal species that produce microsclerotia. Therefore, alternative methods ranging from microplate absorbance assays to droplet-based microfluidics have been developed for filamentous fungi [6, 7]. Beyond screening, novel methods are being

developed to extend microplate reader-based assays for elucidating the mechanism of action [8].

1.3 Fluorescence Microscopy in Antifungal Agent Evaluation

Complementary to the advancements in microplate reader-based assays, fluorescent dye-based microscopic analysis facilitates the cellular visualisation of fungal responses. When filamentous fungi interact with antifungal substances, they display apoptosis-like cell death in addition to necrosis [9]. Staining techniques with DAPI (4',6-diamidino-2-phenylindole), TUNEL (Terminal deoxynucleotidyl transferase dUTP Nick-End Labelling), H2DCFDA (2',7'-dichlorodihydrofluorescein diacetate) and Annexin V are used to illustrate the classic symptoms of programmed cell death [10]. However, these methods are relatively expensive and less suitable for preliminary screening, making dyes such as Propidium iodide and SYTO 13 preferable for quantifying relative cell death. Furthermore, differential staining provides insights into the mechanism of action based on membrane integrity and metabolic activity [11, 12]. This chapter focuses on evaluating the effectiveness of antifungal agents against *M. phaseolina* using protocols designed to ensure reliability.

2. Materials

2.1 Preparation of Fungal Inoculum

1. Potato dextrose agar (PDA)
2. 90 mm sterile disposable Petri plates
3. 20 mL test tubes
4. 1 mL micropipette and tips

5. Parafilm
6. 70 % ethanol
7. Pure PDA culture of *M. phaseolina*
8. Laminar air flow chamber
9. 28°C incubator

2.2 *Antifungal Efficacy Testing*

1. Test AFA and solvent (see **Note 1**)
2. Positive control (PC) and Negative control (PC) (see **Note 2**)
3. 96-well flat-bottom transparent microplates (see **Note 3**)
4. 3-day-old *M. phaseolina* culture plate
5. 2X potato dextrose broth (PDB)
6. 0.2 µm syringe filter and 10 mL syringe (see **Note 4**)
7. 200 µL micropipette and tips
8. Micropore tape
9. 28°C incubator with shaker
10. Microplate reader

2.3 *Fluorescence Microscopy*

1. Propidium iodide – 20 µg/mL (working concentration)

Excitation: ~ 532 nm

Emission: 600–650 nm (red channel)

2. SYTO13 fluorescent dye – 5 µg/mL (working concentration)

Excitation: ~ 488 nm

Emission: 500–540 nm (green channel)

3. Test AFA and solvent (see **Note 1**)
4. 0.2 µm syringe filter and 10 mL syringe
5. 200 µL micropipette and tips
6. Micropore tape
7. PDB
8. Milli-Q® (MQ) water
9. Glass slides and coverslips
10. 96-well flat-bottom transparent microplate
11. 28°C incubator with shaker
12. Confocal microscope

3. Methods

3.1 *M. phasoelina* Agar Culture

1. Autoclave the desired number of test tubes with 10 mL of PDA medium (see **Note 5**).

2. Perform the subsequent steps inside a laminar airflow chamber with all equipment wiped with 70% ethanol.
3. Pour the molten PDA agar into the sterile Petri plates and let it solidify and dry properly.
4. Ensure that the surface of the pure culture on the medium is uniform (see **Note 6**).
5. Punch out the microsclerotia from a fixed radius of the dark-coloured region using a 1 mL micropipette held perpendicular to the agar surface to ensure uniform inoculum (see **Note 7**).
6. Depress the plunger to release the micro-agar plug at the centre of the freshly prepared agar plate. (see **Note 8**).
7. Seal the Petri plate completely with parafilm to avoid contamination and incubate it at 28°C for 3 days.

3.2 Antifungal Efficacy Testing

1. Inoculate the microsclerotia as mentioned in **step 5** of **method 3.1** using a 200 µL micropipette (see **Note 7**).
2. Depress the plunger to release the micro-agar plug into the wells of the microplate (see **Note 8**).
3. Add filter-sterilised AFA to each well, adjust to 100 µL with solvent, and use 100 µL of sterile solutions for the PC and NC treatments (see **Note 9**).
4. Add 100 µL of 2X PDB to each well, wrap the microplate with micropore tape and incubate at 28 °C in a shaker set to 180 rpm (**Fig. 1**).
5. Prepare four replicates of the complete microplate culture for 12-, 24-, 36- and 48 h time points (see **Note 10**).

6. Remove the microplate lid before measuring the optical density (OD) at 595 nm in a microplate reader.
7. Plot OD values vs concentration from each time point and perform non-linear regression analysis (**Fig. 2A**).
8. Select the dose–response curve that most closely resembles an ideal sigmoidal profile, demonstrating the best goodness-of-fit values, a wide dynamic range, and narrow confidence intervals (see **Note 11**).
9. Convert the individual OD values from the selected curve to the percentage of inhibition (PI) using the formula: $PI = \frac{(NC-OD)}{(NC-PC)} \times 100$ (see **Note 12**).
10. Create an XY data table. Enter the concentrations of the AFA into X. Enter the corresponding replicate PI values into Y.
11. From the data table, click Analyse, choose nonlinear regression, choose the panel of equations "Dose-response curves - Inhibition" and then choose the equation "[Inhibitor] vs. Normalised response - Variable slope".
12. Visualise the graph and metadata of IC₅₀ analysis from the respective tabs of the software (**Fig. 2B**).
13. The IC₅₀ values of different AFAs can be compared to evaluate their relative efficacy under the given experimental conditions (see **Note 13**).

3.3 Cell Death Assay

1. Inoculate the microsclerotia into the wells of the microplate as mentioned in **steps 1 and 2 of method 3.2**.

2. Add 200 µL of PDB, wrap the microplate with micropore tape and incubate at 28 °C in a 180 rpm shaker for 24 h. This allows the microsclerotia to germinate and form mycelia (see **Note 14**).
3. Remove PDB using a pipette and wash the germinated microsclerotia thrice with 200 µL of filter-sterilised solvent.
4. Add the calculated volume of sterile AFA for a final well volume of 200 µL, and make up the remaining volume with solvent. Use 200 µL sterile solution of PC and NC for the treatments.
5. Wrap the microplate with micropore tape and incubate at 28 °C in a 180 rpm shaker for 24 h (see **Note 14**).
6. Wash the mycelia three times with MQ and resuspend them in MQ.
7. Transfer the mycelia to a glass slide using a 200 µL cut tip (see **Note 15**).
8. Stain the samples with 10 µL SYTO13 and 10 µL of propidium iodide for 10 min in the dark and place a cover slip on the sample (see **Note 16**).
9. Acquire images of the mycelia using a water-immersion objective and visualise the slide under the confocal microscope (see **Note 17**).
10. Export images that combine the green and red fluorescence channels to represent the total mycelial biomass.
11. Export images showing only the red fluorescence channel (propidium iodide staining) to represent dead mycelial structures.
12. Quantify fluorescence by measuring the area covered by fluorescent pixels in the exported images using calibrated ImageJ software (See **Note 18**).
13. The percentage of cell death (% CD) is calculated by using the formula,

$$\% \text{ CD} = \frac{\text{Red Fluorescence}}{\text{Total Fluorescence}} \times 100.$$

14. Analyse the effect of AFA dosage on the integrity of mycelia as a response parameter (**Fig. 2C**).

4. Notes

1. Solvents without inhibitory effects on the fungus can be used to dissolve multiple AFAs. In this chapter, an antifungal protein dissolved in 1X PBS was used as a representative example.
2. Positive control: a commercially available AFA that shows maximum inhibition.
Negative control: the solvent used for dissolving the AFA.
3. Flat-bottom transparent microplates enable uniform content distribution during incubations and ensure accurate spectrophotometric measurements.
4. Select a syringe filter material compatible with the AFA.
5. Uniform agar plate preparation ensures equal-sized micro-agar plug inoculum.
6. If the culture contains aerial mycelial growth, remove it using an L-rod, and repeat the same procedure for each subculture.
7. Ensure proper insertion of a single micro-agar plug inside the pipette tip during each transfer.
8. Use a fresh tip whenever more than one micro-agar plug is accidentally picked up or when the plug is not being released.
9. For a final volume of 200 μL per well, the volume of AFA added to achieve the desired concentration should not exceed 100 μL .

10. Optimise the AFA concentration range (beginning with 10-fold dilutions followed by 2-fold dilutions), number of technical and biological replicates, and the endpoint readout time based on the phytopathogen under study to obtain a reliable dose–response curve.
11. Statistical analyses were performed using GraphPad Prism version 10.2.0 for Windows (GraphPad Software, Boston, Massachusetts, USA). For further details, visit www.graphpad.com.
12. Experimental variations are minimised by normalising the OD values of each microplate against the corresponding negative and positive control values at the given time point.
13. The variable slope model used here fits the Hill slope from the provided data rather than assuming a standard slope. The software output determines the IC_{50} of the AFA, defined as the concentration that elicits a response halfway between the (maximum) response of the NC and the (maximum inhibition) response of the PC.
14. Select the incubation duration according to the phytopathogen and AFA under study.
15. To prevent mycelial damage during transfers, press the pipette plunger, position the cut tip over the centre of the mycelium, and gently release the plunger until the entire mycelium is drawn inside the tip. Then press the plunger again to transfer.
16. Optimise the concentration of fluorescent dyes based on the developmental stage of the phytopathogen.

17. Fix a magnification between 20X and 60X and repeat the measurement for all the samples.
18. Image analyses were performed using ImageJ for Windows. For further details, visit <https://imagej.net>.

References

1. Marquez N, Giachero ML, Declerck S, Ducasse DA (2021) Macrophomina phaseolina: General Characteristics of Pathogenicity and Methods of Control. *Frontiers in Plant Science* 12. <https://doi.org/10.3389/fpls.2021.634397>
2. Alizadeh M, Khodadadi Manesh S, Fathi P, et al (2025) Biology and Host Ranges of the Plant Pathogenic Fungus Macrophomina Phaseolina: a Comprehensive Review. *Journal of Crop Health* 77:50. <https://doi.org/10.1007/s10343-024-01106-4>
3. Wiederhold NP (2023) Antifungal Susceptibility of Yeasts and Filamentous Fungi by CLSI Broth Microdilution Testing. *Antifungal Drug Resistance: Methods and Protocols* 3–16. https://doi.org/10.1007/978-1-0716-3155-3_1
4. Arendrup MC, Kahlmeter G, Guinea J, Meletiadis J (2021) How to: perform antifungal susceptibility testing of microconidia-forming dermatophytes following the new reference EUCAST method E.Def 11.0, exemplified by *Trichophyton*. *Clinical Microbiology and Infection* 27:55–60. <https://doi.org/10.1016/j.cmi.2020.08.042>

5. Berkow EL, Lockhart SR, Ostrosky-Zeichner L (2020) Antifungal Susceptibility Testing: Current Approaches. *Clinical Microbiology Reviews* 33:10.1128/cmr.00069-19. <https://doi.org/10.1128/cmr.00069-19>
6. McNab E, Rether A, Hsiang T (2023) Development of a microplate absorbance assay for assessing fungicide sensitivity of filamentous fungi and comparison to an amended agar assay. *Journal of Microbiological Methods* 204:106653. <https://doi.org/10.1016/j.mimet.2022.106653>
7. Beneyton T, Wijaya IPM, Postros P, et al (2016) High-throughput screening of filamentous fungi using nanoliter-range droplet-based microfluidics. *Scientific Reports* 6:27223. <https://doi.org/10.1038/srep27223>
8. Wong-Deyrup SW, Song X, Ng T-W, et al (2021) Plant-derived isoquinoline alkaloids that target ergosterol biosynthesis discovered by using a novel antifungal screening tool. *Biomedicine & Pharmacotherapy* 137:111348. <https://doi.org/10.1016/j.biopha.2021.111348>
9. Sharon A, Finkelstein A, Shlezinger N, Hatam I (2009) Fungal apoptosis: function, genes and gene function. *FEMS Microbiology Reviews* 33:833–854. <https://doi.org/10.1111/j.1574-6976.2009.00180.x>
10. Semighini CP, Harris SD (2010) Methods to Detect Apoptotic-Like Cell Death in Filamentous Fungi. *Molecular and Cell Biology Methods for Fungi* 269–279. https://doi.org/10.1007/978-1-60761-611-5_20
11. Van Genechten W, Van Dijck P, Demuyser L (2021) Fluorescent toys ‘n’ tools lighting the way in fungal research. *FEMS Microbiology Reviews* 45:fuab013. <https://doi.org/10.1093/femsre/fuab013>

12. Standaert-Vitse A, Aliouat-Denis C-M, Martinez A, et al (2015) SYTO-13, a Viability Marker as a New Tool to Monitor In Vitro Pharmacodynamic Parameters of Anti-Pneumocystis Drugs. PLOS ONE 10:e0130358. <https://doi.org/10.1371/journal.pone.0130358>

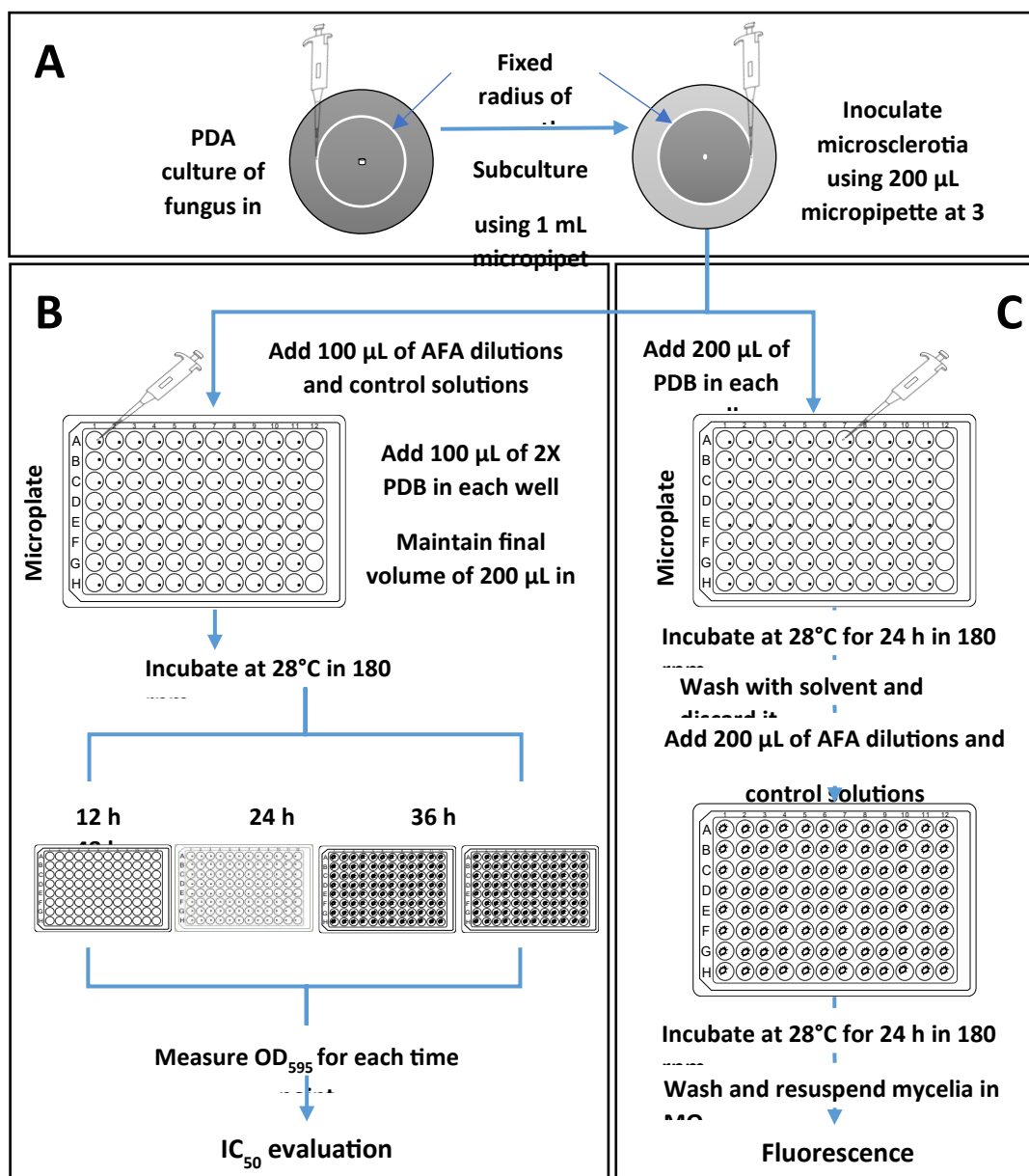


Figure 1. Schematic workflow of antifungal efficacy testing and cell death assay.

- A. Microsclerotia collected from a fixed radius of the agar culture using micropipettes are subcultured and allowed to germinate and form mycelia in subsequent microplate broth culture.
- B. Mycelia are treated with antifungal agents and controls in the presence of PDB. Different time points are used for the spectrophotometric measurements, and one representative time point is selected to evaluate IC₅₀.
- C. Mycelia are incubated in PDB to allow adequate growth. Washing steps, performed both before and after AFA and control treatments, removed residual matter before fluorescence microscopy.

AFA: Antifungal agent, PDA: Potato dextrose agar, PDB: Potato dextrose broth, DAI: Days after inoculation, RPM: Revolutions per minute, OD₅₉₅: Optical density at 595 nm, h: Hours, MQ: Milli Q water, mL: Millilitre, μ L: Microlitre, °C: Degree Celsius, IC₅₀: Inhibitory concentration.

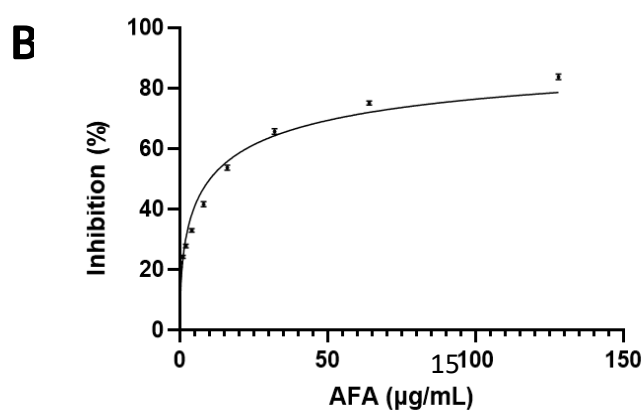
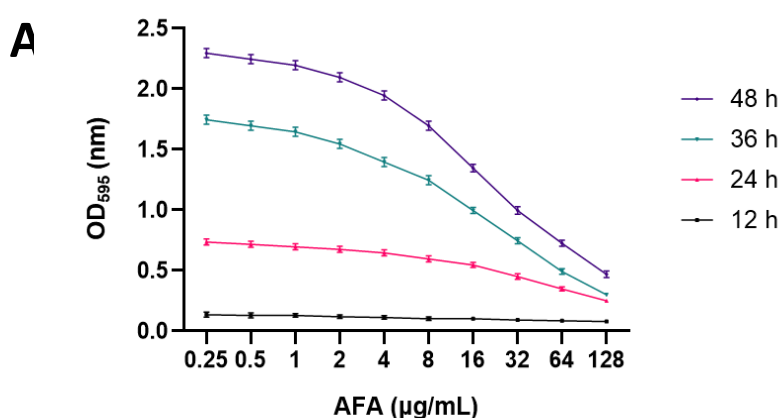


Figure 2. Determination of IC₅₀ value and cell death analysis.

- A. The graph represents mean OD values of the inoculated wells with increasing AFA concentration, measured at four different time points. The error bar indicates SEM with three replicates. The 48 h time point is selected as the statistically reliable dose-response curve.
- B. Dose-response curve represents the inhibitor concentration (AFA) versus normalised response (percentage of Inhibition) at the 48 h time point. The error bar indicates SEM with three replicates. The variable slope model used here fits the Hill slope from the data provided, rather than assuming a standard slope. The software output determines the IC₅₀ of the AFA, which is the concentration that elicits a response halfway between the maximum response of the NC and the maximum inhibition response of the PC.
- C. The response of germinated mycelia after 24h of AFA treatment is represented as the mean percentage of cell death. The error bar indicates SEM with three replicates. One-way ANOVA was performed, and significance was reported at p-value < 0.05.

AFA: Antifungal agent, OD₅₉₅: optical density at 595 nm, PC: Positive control, NC: Negative control, AFA: Antifungal agent, IC₅₀: Inhibitory concentration. %: percentage, mL: Millilitre, µL: Microlitre, µg: Microgram, h: Hours, SEM: Standard error of mean, ANOVA: Analysis of variance.